

Section 7 concludes, while all proofs are provided in the Appendix.

2 Model

There is an underlying state of the world $\theta \in \{L, R\}$, for example, corresponding to whether the left-wing or the right-wing candidate is more qualified for political office. Agents have heterogeneous prior (ideological) beliefs about θ , and agent i 's prior that $\theta = R$ is denoted by b_i with an *ex ante* distribution $H_i(\cdot)$, which may or may not be the same across agents.

Sharing Network. We assume there are N agents in the population, who share a news item according to a *sharing network* defined by matrix $\mathbf{P}$ of link probabilities:

$$\mathbf{P} \equiv \begin{pmatrix} 0 & p_{12} & \cdots & p_{1N} \\ p_{21} & 0 & \cdots & p_{2N} \\ \cdots & \cdots & \cdots & \cdots \\ p_{N1} & p_{N2} & \cdots & 0 \end{pmatrix}$$

where p_{ij} is the probability that agent i has a link to agent j . We define agent i 's neighborhood $\mathcal{N}_i$ as the set of agents attached to her with an outgoing link, and denote her degree or the size of her neighborhood by $|\mathcal{N}_i|$. The sharing network reflects both an individual's social circle and the algorithms the platform uses for promoting shared content. The news item in question could be a news article or a post by one of the users, and throughout we refer to it as an "article".

Misinformation and News Generation. Each article has a three-dimensional type (r, m, ν) . Here, $r \in [0, 1]$ indicates the *reliability* of the news, and $m \in \{L, R\}$ is the *message*, which corresponds to the article's viewpoint, for example, whether it argues for a left-wing or right-wing idea. Finally, ν is the article's *veracity*, which can either be $\mathcal{T}$, to indicate the article is truthful, or $\mathcal{M}$, to indicate the article contains *misinformation*.⁸

We assume that, at the beginning of the game, the type vector (r, ν, m) is drawn according to the following i.i.d. process:

- (i) The article's reliability $r \in [0, 1]$ is drawn from a continuous distribution F with density f .
- (ii) The veracity of the article is $\nu = \mathcal{T}$ (contains truthful content) with probability $\phi(r)$ or is $\nu = \mathcal{M}$ (contains misinformation) with probability $1 - \phi(r)$. We assume that ϕ is increasing and differentiable in r , and satisfies $\phi(0) = 0$ and $\phi(1) = 1$, so that the least reliable article always contains misinformation, and as the degree of reliability increases, the likelihood of misinformation monotonically declines and reaches zero.

⁸Our focus in this paper is on misinformation, interpreted as items containing misleading information or arguments that can influence (a subset of) the public. Articles containing misinformation are in practice much more numerous than those that can be classified as "fake news", which explicitly propagate demonstrably false information (e.g., [Egelhofer and Lecheler \(2019\)](#), [Allen et al. \(2020\)](#), [Guess et al. \(2019\)](#), [Grinberg et al. \(2019\)](#)). For example, according to this definition a news item that favorably describes a report denying climate change, without putting this in the context of hundreds of other reports reaching the opposite conclusion or mentioning the criticisms that it has received from experts, contains misinformation.

- (iii) If $\nu = \mathcal{T}$ (the article is truthful), then its message is generated as $m = \theta$ with probability $p > 1/2$. Conversely, if $\nu = \mathcal{M}$ (the article contains misinformation), then its message is generated as $m = \theta$ with probability $q \leq 1/2$ and is weakly anti-correlated with the truth.

While m and r are common knowledge (for example, the message m is directly observed and reliability depends on certain commonly-observed characteristics such as source and headline), the third dimension, ν , is unknown to all agents. We assume that agents update their beliefs about ν using Bayes' rule given beliefs about the underlying state θ and the observables (r, m) of the article.

Social Media Behavior. Time is discrete $t = 1, 2, \dots$ Upon receipt of the article, an agent i can take one of three actions $a_i \in \{\mathcal{S}, \mathcal{I}, \mathcal{D}\}$, as described below:

- (i) Share ($\mathcal{S}$): The agent decides to *share* the article and passes it onto others after her.
- (ii) Ignore ($\mathcal{I}$): The agent decides to *ignore* the article and does not engage with it.
- (iii) Dislike ($\mathcal{D}$): The agent decides to *dislike* the article, which means expressing disagreement or contempt for the content contained in it.

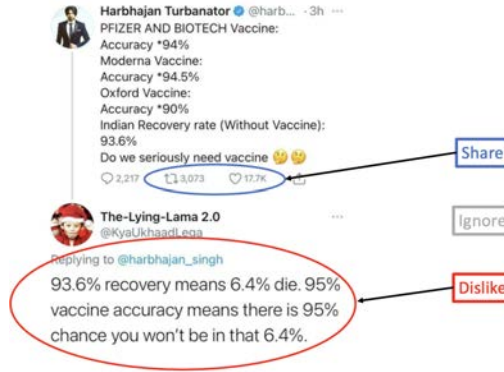


Figure 1. Sample Tweet.

The three possible actions are depicted in Figure 1 using Twitter as a sample social media platform. A given user sees the article and decides how to respond to it. She can (i) share it ($\mathcal{S}$), which actively puts it on other social media news feeds; (ii) ignore it ($\mathcal{I}$), where the user simply scrolls past the article; or (iii) actively dislike it ($\mathcal{D}$), expressing derision for the content.

At time $t = 1$, we assume that some initial seed agent i^* begins with the article. If the article is shared by agent i , it is passed to all $j \in \mathcal{N}_i$. In contrast, following ignore or dislike, the article does not propagate past agent i .

Payoffs. Let us define shares after i as $S_i = |\{j \in \mathcal{N}_i : a_j = \mathcal{S}\}|$ and dislikes after i as $D_i = |\{j \in \mathcal{N}_i : a_j = \mathcal{D}\}|$. Agent i 's utility can then be written as

$$U_i = \begin{cases} 0, & \text{if } a_i = \mathcal{I} \\ \tilde{u}1_{\nu=\mathcal{M}} - \tilde{c}, & \text{if } a_i = \mathcal{D} \\ u1_{\nu=\mathcal{T}} - c1_{\nu=\mathcal{M}} + \kappa S_i - dD_i, & \text{if } a_i = \mathcal{S} \end{cases} \quad (1)$$

where $\mathbf{1}$ is the indicator function (equal to 1 if true and 0 otherwise). Here, $\tilde{u}$, $\tilde{c}$, u , c , κ and d are strictly positive parameters, which we discuss below.

- (i) We normalize payoffs following ignore, $\mathcal{I}$, to $U_i = 0$.
- (ii) Payoffs from dislike, $\mathcal{D}$, depend on whether the article contains misinformation. We assume, in particular, that disliking has a cost of $\tilde{c} > 0$, regardless of whether the article is truthful (because of, say, the effort required to actively call out misinformation). In addition, disliking an article containing misinformation has a benefit of $\tilde{u} > \tilde{c}$, because individuals like calling out misleading articles. This formulation implies that disliking is never preferred to ignoring for an article that is truthful with probability 1, and is always preferred to ignoring for an article that contains misinformation with probability 1.
- (iii) Following a decision to share, $\mathcal{S}$, an agent receives utility from two sources. First, agents receive utility from sharing truthful content, but incur a cost from sharing misinformation. This explains the first component of utility following $\mathcal{S}$, $U_i^{(1)} = u\mathbf{1}_{\nu=\mathcal{T}} - c\mathbf{1}_{\nu=\mathcal{M}}$. Second, agents enjoy positive feedback from their peers (such as likes, or in our setting re-shares), but are negatively affected by dislikes. This is captured by the second component of utility $U_i^{(2)} = \kappa S_i - dD_i$. In this formulation, the parameter κ captures the importance of “popularity” for the agent’s sharing decision, while $-d$ represents the extent to which she cares about negative reactions.⁹

Information Structure and Solution Concept. Agents are not aware of when the article was first introduced onto social media, the prior sharing process, or the structure of the social network (though the link matrix $\mathbf{P}$ is common knowledge).¹⁰ Moreover, while any agent i knows the distribution $\{H_i\}_{i=1}^N$ of beliefs in the population, she does not know any agent j ’s belief (ideology) b_j . We focus on Bayesian-Nash equilibria, and refer to these as “equilibria” for short.

To eliminate trivial and unrealistic equilibria, we assume that the sensationalism of an article is upper bounded by $\bar{\kappa} = (c\tilde{c} - u(\tilde{u} - \tilde{c})) / (\tilde{u}N)$. This assumption guarantees that there is never an equilibrium where *every* agent *always* shares *all* articles. It also eliminates equilibria where agents may share and dislike, but never ignore.

Discussion — The basic assumptions introduced above are consistent with salient patterns of behavior and information structure in social media. First, as documented in studies such as Pennycook et al. (2021), users want to share content they believe to be truthful and not contain misinformation. Second, while users derive value from peer encouragement and re-shares on social media (Eckles et al. (2016)), they also suffer reputational costs when they get called out for sharing misinformation (see,

⁹Reputational loss associated with dislikes can be micro-founded straightforwardly by assuming that there are two types, normal and careless, and normal agents would like to distinguish themselves from careless ones who always share. We do not pursue this path, since imposing exogenous reputation costs simplifies the notation and analysis, with no major cost in terms of insights.

¹⁰More formally, we assume that all agents have uniform priors over these objects. In this sense, agents do not know the exact interactions and sharing patterns outside their neighborhood, which is consistent with the evidence in Breza et al. (2018). That being said, because the equilibrium sharing process is Markovian, this assumption can be relaxed by replacing $\mathbf{P}$ with the adjacency matrix.

for example, evidence from Facebook in [Altay et al. \(2020\)](#)). Finally, social media users often engage in criticisms of available content and inform others about misinformation (see, for example, [Kim et al. \(2020\)](#) for evidence in the context of 2018 midterm elections).

3 Equilibria in General Networks

In this section, we characterize the structure of equilibria for any sharing network structure $\mathbf{P}$ and provide various comparative statics. Without loss of generality (and ease of exposition), we fix the article's message as $m = R$ for the remainder of the paper.¹¹

3.1 Cutoff Strategies and Strategic Complementarities

When agent i receives an article with reliability r and message $m = R$, she updates her (*ex post*) belief, π_i , that the article is truthful according to Bayes' rule:

$$\pi_i = \frac{(pb_i + (1-p)(1-b_i))\phi(r)}{(qb_i + (1-q)(1-b_i))(1-\phi(r)) + (pb_i + (1-p)(1-b_i))\phi(r)}. \quad (2)$$

Clearly, π_i is increasing in b_i since an agent is more likely to believe in an article's veracity when its message agrees with her prior. Moreover, π_i is increasing in r , as the agent updates more on the basis of more reliable articles.

We can also see that the payoff to sharing ($\mathcal{S}$) increases in π_i , since the first component of utility, $U_i^{(1)}$, is increasing in π_i (as the individual would like to share truthful articles), while $U_i^{(2)}$ is independent of π_i . With a similar reasoning, the payoff to disliking ($\mathcal{D}$) is decreasing in π_i , whereas the payoff to ignoring ($\mathcal{I}$) is independent of π_i . This monotone behavior of payoffs will lead to best-response decision rules for agents that take the form of cutoff strategies, as we explain next.

We say that agent i employs a *cutoff strategy* if there exists $b_i^*(r)$ and $b_i^{**}(r)$ such that agent i chooses $\mathcal{S}$ when $b_i > b_i^{**}(r)$, chooses $\mathcal{I}$ when $b_i^*(r) < b_i < b_i^{**}(r)$, and chooses $\mathcal{D}$ when $b_i < b_i^*(r)$. Cutoff strategies in our context imply that agents who strongly agree with an article tend to share it, agents who strongly disagree with it tend to choose dislike, and those with intermediate beliefs typically ignore the article.

We will see in the next theorem that all equilibria will be in cutoff strategies. This means, in particular, that an equilibrium can be summarized by cutoff vectors $(\mathbf{b}^*, \mathbf{b}^{**}) = (b_1^*, b_1^{**}, \dots, b_N^*, b_N^{**})$. Furthermore, these cutoffs $b_i^*(r)$ and $b_i^{**}(r)$ will both be decreasing in r , so that as reliability increases, an article becomes more likely to be shared and less likely to be disliked.

We can also note that our social media game exhibits *strategic complementarities*. To see this, observe that when others share more, meaning that b_i^{**} (weakly) decreases for all i , this tends to encourage more sharing for every agent j , because the second component of utility, $U_j^{(2)}$, increases, raising the overall utility of sharing. Similarly, when others reduce their likelihood of disliking, meaning that now b_i^* (weakly) decreases for all i , this reduces the likely cost of sharing misinformation by

¹¹To see this is without loss of generality, observe that the analysis applies identically with an $m = L$ message but with complementary priors $b_i' = 1 - b_i$.